Shoumik Saha

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RESEARCH INTEREST

- **Security** - Computer security, Malware, Exploits, Adversarial Attack
- **Machine Learning** - Adversarial Machine Learning, Machine Learning for Security

PROFESSIONAL EXPERIENCE

Graduate Research Assistant **Full Time**
Department of Computer Science *Aug 2022 – Present*
University of Maryland

Supervisor - Dr. Tudor Dumitras

Lecturer **Full Time**
Department of Computer Science and Engineering *Jul 2021 – Jul 2022*
United International University

Research Assistant **Part Time**
Data Science and Engineering Research Lab *Mar 2021 – Jul 2022*
Bangladesh University of Engineering and Technology (BUET)

Under the supervision of
Dr. Mohammed Eunus Ali, Dr. Atif Hasan Rahman

EDUCATION

University of Maryland
Ph.D. in Computer Science and Engineering (Ongoing) *Aug 2022 – Present*

- CGPA – *3.85/4.0*
- Advisor – Dr. Tudor Dumitras

Bangladesh University of Engineering and Technology (BUET)
BSc. in Computer Science and Engineering *Mar 2016 – Feb 2021*

- CGPA – *3.66/4.0*
- Advanced GPA – *3.79/4.0*
- Thesis supervisor – Dr. Atif Hasan Rahman

Notre Dame College
Higher Secondary Certificate *Jul 2013 – Jun 2015*

- GPA – 5.00/5.00

RESEARCH

- **Shoumik Saha**, Sadia Afroz, Atif Rahman.
  MALIGN: Adversarially Robust Malware Family Detection using Sequence Alignment (BSc Thesis) (arXiv) (pdf) (Updated Version)
  A novel approach that incorporated ideas and tools from Bioinformatics into Malware Security. We developed a static raw-byte-based malware family classifier with better accuracy and robustness. Malign can give insight to defenders by relocating the code blocks responsible for malicious attacks. **Status:** Proceeded to the rebuttal period of IEEE S&P 2022 (acceptance rate: 14.9%)
- **Shoumik Saha**, Wenxiao Wang, Yigitcan Kaya, Soheil Feizi.
  Adversarial Robustness of Learning-based Static Malware Classifiers (arxiv) (pdf)
  We explored Universal Append Patch (UAP) attack on byte-based static malware classifier for the first time. Then we applied de-randomized smoothing on our defense model to make it robust against such attack. **Status:** Ongoing

SCHOLARSHIPS AND ACHIEVEMENTS

- Awarded *'Dean's Fellowship'* from the University of Maryland
- Awarded *'Innovation Fund'* for research from the ICT division of Bangladesh government
- Achieved *'Merit Stipends'* from BUET in five out of seven terms
- Got *'Dean's Award'* in Junior year for extra-ordinary result
- Achieved *'Talent-Pool Scholarship'* in High School

SELECTED ACADEMIC PROJECTS

- **Website Development for Flight and Hotel Room Reservation System** (Github link)
  Developed a website where customers can book flight tickets, hotel rooms through card payment, and companies can update their page and manage bookings. *Tools:* Django, SQLite
- **Software Development for Football Club Management** (Github link)
  This application stores all information about the club (players, managers, games). Admin of the club can update, add and delete these information and generate graphs from statistics. *Tools:* Java, JavaFX, Oracle Database
- **Micro-controller Project for Real-time Detection of Car Theft** (Youtube video link)
  Can detect a car in real time and send a warning message with gps location to the owner's mobile. *Tools:* Arduino, GPS, GSM module, Sonar, LDR sensor
- **Multi-player Shooting Game** (Github link)
  A 2nd year project, two players can connect from two different machines under the same network. *Tools:* Java, JavaFX, Java Networking
- **Tic-Tac-Toe** (Github link)
  A 1st year project, single player Tic-Tac-Toe game with Mini-Max algorithm. *Tools:* C, iGraphics
- **Miscellaneous**
  - Developing a C compiler using lexical analyzer and parser designing tools
  - Designing a 4-bit Computer Model using Atmel Studio, MIPS architecture
  - Simulating Mancala game in AI lab course
  - Implementing and modifying some functionalities of XV6 Operating System
  - Modifying some functionalities of Computer Network in NS2.

COURSES INSTRUCTED

- Discrete Mathematics
- Data Structure and Algorithm
- Operating Systems
- Digital Logic Design

ACTIVITIES

- General Secretary, BUET Photographic Society
- President, Notre Dame Nature Study Society
- Participant, ICPC Dhaka Regional
- Vice President, Gregorian Science Club

STANDARDIZED TEST SCORES

- *TOEFL:* 111/120 (Reading: 28, Listening: 28, Speaking: 27, Writing: 28)
- *GRE:* 321/340 (Quant: 167, Verbal: 154, AWA: 3.5)